PALAMAKULA DEEPIKA

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Professional Summary

AI/ML Engineer with expertise in the full data lifecycle, specializing in data preprocessing, model validation, and advanced visualization. Technically proficient in Python, AWS Cloud deployment, and IoT system development via Arduino IDE. Proven ability to transform complex datasets into actionable business intelligence using Power BI. Experienced deploying MLOps pipelines on AWS. Proven track record of increasing model accuracy by 10% and reducing manual data prep by 30% through automation.

Education

Bachelors of technology in Artificial Intelligence & Machine Learning | 2021 – 2025
Saveetha Engineering College | Chennai, TamilNadu, India | CGPA 8.2/10.0

Technical Skills

- Programming Languages: Python, Java, C, SQL (MySQL), Oracle SQL.
- Machine Learning & Data: Data Structures & Algorithms (DSA), Pandas, NumPy, Scikit-learn, XGBoost.
- Databases & Storage: SQL Server, NoSQL (MongoDB), Oracle Database Management.
- Cloud & DevOps: Google Cloud GCP, DevOps Fundamentals, GitHub, AWS (EC2, S3), Git, Docker
- Data Visualization: Power BI, Microsoft Excel (Advanced), Interactive Data Dashboards.
- Design & Prototyping: UI/UX Design, Figma, Wireframing, Power BI, Arduino IDE.
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Work Experience

Machine Learning Intern | FabHost Complete Web Sol. | *Jan 2024 – Mar 2024*

- Developed and deployed supervised machine learning models using Python to solve complex predictive modeling challenges, achieved a 92% Precision/Recall score using XGBoost and validated results with SHAP values, data-driven insights.
- Engineered end-to-end data pipelines, including Feature Engineering and Model Validation, which improved algorithmic stability and model robustness across various datasets.
- Automated data cleaning and integration processes by developing custom Python scripts, significantly reducing manual data preparation time and streamlining the ML Lifecycle (MLOps).

Projects

CareerLens AI: Scalable Career Recommendation System | *Python, FastAPI, Flutter, AWS*

- Architected a 3-layer AI system (Flutter, FastAPI, Python ML) to provide personalized career path recommendations and skill gap analysis with 90%+ predictive accuracy.
- Integrated Explainable AI (SHAP) with RandomForest/XGBoost models to offer transparent, feature-level reasoning and confidence scores for user recommendations.
- Deployed a cloud-ready infrastructure on AWS EC2 and S3 using MongoDB for scalable data storage, ensuring modular design and seamless RESTful API communication.

Toxic Gas Leak Detection & Alerting Device | *IoT, ESP32, Random Forest, IFTTT*

- Developed an IoT safety system utilizing ESP32 and MQ135 sensors to monitor atmospheric toxicity and deliver real-time SMS/Email alerts via IFTTT protocols.
- Implemented Random Forest Regression (RFR) to analyze sensor telemetry and predict hazard levels, reducing potential emergency response time through automated detection.
- Engineered a cloud-integrated monitoring solution using Arduino IDE and Blynk, enabling remote device control and live data visualization of environmental metrics.

Certifications

- Data Structures & Algorithms (C & Java) | Udemy | 2023
- Fundamentals of DevOps | Simplilearn | 2024
- AWS Certified Cloud Practitioner | Amazon Web Services (AWS) | 2025